

Machine Learning

Tutorial Question Bank — Full Solutions

20 Questions · Exam Preparation Guide

Topics: Logistic Regression · Naïve Bayes · GDA · PCA · K-Means · Decision Trees

Includes: Intuition · Step-by-Step Workings · Formulas · Revision Checklist

This document provides complete model answers to all 20 tutorial questions. Each answer begins with an intuition summary, followed by step-by-step working with explicit formulas, and concludes with the final answer. A revision checklist appears at the end.

Q1. Discriminative vs Generative Classifiers

Logistic regression is sometimes described as a discriminative classifier, while GDA is a generative classifier. Explain the distinction, when GDA outperforms logistic regression, and when logistic regression is preferred.

■ Intuition

A generative model learns the full joint distribution $P(x, y) = P(x|y) \cdot P(y)$, so it can 'generate' new samples. A discriminative model directly models $P(y|x)$, the decision boundary, ignoring how x was produced.

■ Generative — GDA

Models $P(x | y=0)$ and $P(x | y=1)$ as Gaussians plus class prior $P(y)$. Uses Bayes rule: $P(y|x) = P(x|y) \cdot P(y) / P(x)$.

■ Discriminative — Logistic Regression

Directly models $P(y=1 | x) = \sigma(\theta^T x)$ without any assumption about $P(x)$.

■ When GDA Wins

- Data is truly Gaussian — GDA's stronger assumption is correct → lower variance, better performance with small n .
- Training data is limited — extra structure helps.
- Features are continuous and roughly Gaussian.

■ When Logistic Regression Wins

- Gaussian assumption is violated (e.g., binary/count features).
- Large dataset — can afford to estimate $P(y|x)$ from data alone.
- GDA is a special case: if GDA assumptions hold, its decision boundary is logistic, so LR converges to the same boundary asymptotically but is more robust when assumptions fail.

✓ **GDA = generative (models $P(x|y)$); LR = discriminative (models $P(y|x)$ directly). GDA wins with small, Gaussian data; LR wins when assumptions are violated.**

Q2. Naïve Bayes — Zero Probability & Laplace Smoothing

A Naïve Bayes model is trained where 'offer' never appears in legitimate email. Describe what goes wrong at test time and how Laplace smoothing resolves it.

■ Intuition

Naïve Bayes multiplies per-feature likelihoods. A zero probability for any feature kills the entire product — the class is assigned probability zero regardless of all other evidence.

■ What Goes Wrong

Raw MLE estimate: $P(\text{word}=\text{'offer'} | y=0) = 0/N_0 = 0$.

At test time for a legitimate email containing 'offer':

$$P(y=0 | x) \propto P(y=0) \cdot \prod P(x_i | y=0)$$

The term $P(\text{offer} | y=0) = 0$ makes the whole product zero, so the classifier always predicts spam — a zero-probability catastrophe.

■ Laplace (Add-1) Smoothing

Add a pseudo-count of 1 to every word count for every class:

$$P(\text{word } w | y=k) = (\text{count}(w, k) + 1) / (N_k + |V|)$$

where:

- $\text{count}(w, k)$ = number of times word w appears in class- k emails
- N_k = total word tokens in class k
- $|V|$ = vocabulary size

Now even unseen words get a small non-zero probability, eliminating the zero-product problem.

✓ Laplace smoothing adds 1 to every count, ensuring $P(\text{word}|\text{class}) > 0$ always.

Q3. Logistic Regression — Linear Boundary & Feature Engineering

A colleague claims logistic regression always produces a straight-line boundary. Is this correct? How can non-linear boundaries be achieved?

■ Intuition

The claim is partially correct — LR is linear in the feature space it receives. But you can feed it non-linear features.

■ Why the Boundary Is Linear in x

LR predicts $y=1$ when $\theta_0 + \theta_1 x_1 \geq 0$. This is a hyperplane in x -space.

Decision boundary: $\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$ (a line in 2-D)

■ Overcoming the Limitation — Feature Mapping

Replace x with a non-linear feature vector $\phi(x)$. The boundary is linear in $\phi(x)$ but non-linear in original x .

Example — polynomial features in 2-D:

$$\phi(x) = [1, x_1, x_2, x_1^2, x_1 x_2, x_2^2]$$

With these 6 features, LR can learn ellipses, parabolas, and other conic decision boundaries in the original space.

■ Other Approaches

Kernel methods (kernel logistic regression) implicitly map to infinite-dimensional feature spaces. Neural networks learn the feature mapping end-to-end.

✓ The claim is true for raw features. Polynomial (or other) feature mapping lets LR model non-linear boundaries without changing the classifier.

Q4. Naïve Bayes — Full Worked Calculation

Training set: (1,1,1), (1,0,0), (0,1,0), (0,0,0). Compute $P(y=1 \mid x_1=1, x_2=0)$ using Naïve Bayes with binary features.

■ Step 1 — Class Priors

4 examples total: $y=1$ once, $y=0$ three times.

$$P(y=1) = 1/4 \quad P(y=0) = 3/4$$

■ Step 2 — Likelihoods from Training Data

For $y=1$ (only example: $x_1=1, x_2=0$):

$$P(x_1=1 \mid y=1) = 1/1 = 1$$

$$P(x_2=0 \mid y=1) = 0/1 = 0$$

For $y=0$ (examples: (1,0), (0,1), (0,0)):

$$P(x_1=1 \mid y=0) = 1/3$$

$$P(x_2=0 \mid y=0) = 2/3$$

■ Step 3 — Apply Naïve Bayes

$$P(y=1 \mid x_1=1, x_2=0) \propto P(y=1) \cdot P(x_1=1 \mid y=1) \cdot P(x_2=0 \mid y=1)$$

$$= (1/4) \cdot 1 \cdot 0 = 0$$

$$P(y=0 \mid x_1=1, x_2=0) \propto P(y=0) \cdot P(x_1=1 \mid y=0) \cdot P(x_2=0 \mid y=0)$$

$$= (3/4) \cdot (1/3) \cdot (2/3) = 6/36 = 1/6$$

■ Step 4 — Normalise

Total unnormalised = 0 + 1/6.

$$P(y=1 \mid x_1=1, x_2=0) = 0 / (1/6) = 0$$

$$P(y=0 \mid x_1=1, x_2=0) = 1$$

✓ $P(y=1 \mid x_1=1, x_2=0) = 0 \rightarrow$ zero probability because $x_2=0$ was never seen with $y=1$ (Laplace smoothing would fix this).

Q5. PCA — Projection, Reconstruction & Error

$x = [4, 2]^T$, $u_1 = (1/\sqrt{5})[2, 1]^T$. Compute (a) projection score z , (b) reconstructed point $x^\wedge$, (c) reconstruction error $\|x - x^\wedge\|^2$.

■ (a) Projection Score $z = u_1^T x$

$$u_1 = [2/\sqrt{5}, 1/\sqrt{5}]^T$$

$$z = u_1^T x = (2/\sqrt{5}) \cdot 4 + (1/\sqrt{5}) \cdot 2 = 8/\sqrt{5} + 2/\sqrt{5} = 10/\sqrt{5} = 2\sqrt{5}$$

$$z \approx 4.472$$

■ (b) Reconstructed Point $x^\wedge = z \cdot u_1$

$$x^\wedge = z \cdot u_1 = 2\sqrt{5} \cdot (1/\sqrt{5}) \cdot [2, 1]^T = 2 \cdot [2, 1]^T = [4, 2]^T$$

■ (c) Reconstruction Error $\|x - x^\wedge\|^2$

$$x - x^\wedge = [4, 2]^T - [4, 2]^T = [0, 0]^T$$

$$\|x - x^\wedge\|^2 = 0$$

The reconstruction error is zero because x happens to lie exactly on the direction of u_1 — it has no component orthogonal to the first PC.

✓ $z = 2\sqrt{5} \approx 4.47$, $x^\wedge = [4, 2]^T$, Reconstruction error = 0.

Q6. K-Means — Failure Modes & Remedies

Give two distinct failure modes of K-means and a concrete remedy for each.

■ Failure Mode 1 — Sensitivity to Initialisation

K-means can converge to a poor local minimum depending on the starting centroids, missing the globally optimal clustering.

Remedy: K-Means++ initialisation — choose centroids sequentially, weighting each candidate by its squared distance from the nearest existing centroid:

$$P(x \text{ chosen as next centroid}) \propto \min_j \|x - \mu_j\|^2$$

This spreads initial centroids and dramatically reduces poor convergence.

■ Failure Mode 2 — Assumes Spherical, Equal-Size Clusters

K-means minimises within-cluster sum of squared distances, implicitly assuming round clusters of similar density. It fails on elongated, irregular, or very different-sized clusters.

Remedy: Use Gaussian Mixture Models (EM algorithm) which allow clusters with arbitrary covariance (shape) and unequal mixing weights (size). Alternatively, use DBSCAN for arbitrary-shape clusters.

✓ (1) Bad init $\rightarrow$ K-Means++. (2) Non-spherical clusters $\rightarrow$ GMM or DBSCAN.

Q7. K-Means — Elbow Method

Describe the elbow method. What is plotted? Why does it always decrease? What does the elbow indicate?

■ Quantity Plotted

The Within-Cluster Sum of Squares (WCSS), also called inertia:

$$\text{WCSS}(K) = \sum_k \sum_{\{x \in C_k\}} \|x - \mu_k\|^2$$

is plotted on the y-axis against K (number of clusters) on the x-axis.

■ Why WCSS Always Decreases

Adding an extra cluster can only split an existing cluster, reducing or at worst maintaining WCSS. It is strictly non-increasing in K. At K = n, every point is its own cluster and WCSS = 0.

■ The Elbow

For small K, adding one more cluster gives a large WCSS drop. Eventually the marginal gain flattens. The elbow is the point of diminishing returns — a 'kink' in the curve — and is taken as the optimal K. Beyond the elbow, extra clusters explain little variance.

✓ **Plot WCSS vs K; WCSS always falls because more clusters always partition better. The elbow = where marginal improvement becomes small = best K.**

Q8. Imbalanced Data — Beyond Accuracy

A classifier on 95/5 imbalanced data achieves 95% accuracy by predicting negative for every example. Why is accuracy misleading? Name two better metrics.

■ Why Accuracy Misleads

Accuracy counts all correct predictions equally:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

A trivial all-negative classifier gets $950/1000 = 95\%$ accuracy despite catching zero positive cases ($\text{TP} = 0$). It appears excellent while being completely useless for the minority class.

■ Better Metric 1 — Precision & Recall (and F1)

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \text{ [of predicted positives, how many are real?]}$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \text{ [of real positives, how many did we catch?]}$$

$$\text{F1 Score} = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$$

For the all-negative classifier: $\text{Recall} = 0/50 = 0$, so $\text{F1} = 0$. Correctly penalises the useless model.

■ Better Metric 2 — AUC-ROC

Area Under the ROC Curve (TPR vs FPR at all thresholds). $\text{AUC} = 0.5$ for a random classifier; the all-negative model has $\text{AUC} = 0.5$, not 0.95, exposing its failure.

✓ **Accuracy ignores class distribution. F1-score or AUC-ROC correctly evaluate performance on the minority class.**

Q9. PCA — Variance Explained

Eigenvalues: $\lambda_1=60, \lambda_2=40, \lambda_3=25, \lambda_4=15$, remaining 196 sum to 60. (a) Total variance. (b) Fraction explained by PC1–PC3. (c) Components for 70%.

■ (a) Total Variance

$$\text{Total} = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \Sigma \text{ rest} = 60 + 40 + 25 + 15 + 60 = 200$$

■ (b) Fraction Explained by First 3 PCs

$$\begin{aligned} \text{Fraction} &= (\lambda_1 + \lambda_2 + \lambda_3) / \text{Total} = (60 + 40 + 25) / 200 = 125/200 \\ &= 0.625 = 62.5\% \end{aligned}$$

■ (c) Components Needed for $\geq 70\%$

Cumulative variance:

$$k=1: 60/200 = 30.0\%$$

$$k=2: 100/200 = 50.0\%$$

$$k=3: 125/200 = 62.5\%$$

$$k=4: 140/200 = 70.0\% \checkmark$$

✓ (a) 200 (b) 62.5% (c) $k = 4$ components needed.

Q10. K-Means — One Complete Iteration

Dataset $\{3, 5, 11, 13\}$, $K=2$. Initial centroids $\mu_1=4, \mu_2=12$. Carry out one iteration. Has it converged?

■ Step 1 — Assignment

Assign each point to nearest centroid:

Point 3: $|3-4|=1$, $|3-12|=9 \rightarrow C1$

Point 5: $|5-4|=1$, $|5-12|=7 \rightarrow C1$

Point 11: $|11-4|=7$, $|11-12|=1 \rightarrow C2$

Point 13: $|13-4|=9$, $|13-12|=1 \rightarrow C2$

$C1 = \{3, 5\}$, $C2 = \{11, 13\}$

■ Step 2 — Update Centroids

$$\mu1_new = (3 + 5) / 2 = 4$$

$$\mu2_new = (11 + 13) / 2 = 12$$

■ Convergence Check

$\mu1_new = 4 = \mu1_old$ and $\mu2_new = 12 = \mu2_old$. No change!

✓ **Algorithm has converged after 1 iteration. Final clusters: $C1=\{3,5\}$, $C2=\{11,13\}$.**

Q11. Decision Tree — Gini Impurity

12 examples: 7 class A, 5 class B. Split $\rightarrow$ left: 6A, right: 1A+5B. Compute (a) parent Gini, (b) weighted post-split Gini, (c) reduction.

■ Formula

$$\text{Gini}(\text{node}) = 1 - \sum_k p_k^2$$

■ (a) Parent Gini

$$p_A = 7/12, p_B = 5/12$$

$$\text{Gini}(\text{parent}) = 1 - (7/12)^2 - (5/12)^2$$

$$= 1 - 49/144 - 25/144 = 70/144 \approx 0.486$$

■ (b) Weighted Gini After Split

Left child (6 examples, all A): $p_A=1$, $p_B=0$

$$\text{Gini}(\text{left}) = 1 - 1^2 - 0^2 = 0 \text{ (pure!)}$$

Right child (6 examples, 1A+5B): $p_A=1/6$, $p_B=5/6$

$$\text{Gini}(\text{right}) = 1 - (1/6)^2 - (5/6)^2 = 1 - 1/36 - 25/36 = 10/36 \approx 0.278$$

Weighted sum (equal sizes, 6+6=12):

$$\text{Gini}(\text{split}) = (6/12) \cdot 0 + (6/12) \cdot 0.278 = 0.139$$

■ (c) Impurity Reduction

$$\Delta \text{Gini} = 0.486 - 0.139 = 0.347$$

✓ **Large reduction of 0.347 (from 0.486 to 0.139) $\rightarrow$ this is an excellent split.**

Q12. Decision Tree — Overfitting

A tree achieves 100% train accuracy but 60% test accuracy. Name this, and give two strategies to fix it.

■ Phenomenon

This is overfitting — the tree has memorised the training data, learning noise and outliers rather than the true underlying pattern. The large train-test gap (100% vs 60%) is the key indicator.

■ Strategy 1 — Pre-Pruning (before full growth)

Stop growing the tree early by imposing constraints:

- `max_depth`: limit tree depth.
- `min_samples_split`: require a minimum number of examples to split a node.
- `min_impurity_decrease`: only split if reduction $\geq$ threshold.

These hyperparameters are tuned via cross-validation.

■ Strategy 2 — Post-Pruning (after full growth)

Grow the tree fully, then remove branches that provide little test-set improvement:

- Cost-complexity (α) pruning: penalise tree size, remove subtrees whose removal increases error by less than α per leaf.
- Reduced-error pruning: collapse subtrees if accuracy on a validation set does not decrease.

✓ **Overfitting. Fix with pre-pruning (depth/size limits) or post-pruning (cost-complexity pruning).**

Q13. GDA — Maximum Likelihood Estimators

Write the closed-form MLE for ϕ , μ_0 , μ_1 , Σ in GDA with shared covariance.

■ Setup

n training examples, m = number with $y=1$, $(n-m)$ with $y=0$.

■ Class Prior $\phi = P(y=1)$

$$\phi = (1/n) \sum_i 1\{y^{(i)} = 1\} = m / n$$

■ Class Means μ_0 , μ_1

$$\mu_0 = [\sum_i 1\{y^{(i)}=0\} x^{(i)}] / \sum_i 1\{y^{(i)}=0\}$$

$$\mu_1 = [\sum_i 1\{y^{(i)}=1\} x^{(i)}] / \sum_i 1\{y^{(i)}=1\}$$

i.e., the sample mean of each class.

■ Shared Covariance Σ

$$\Sigma = (1/n) \sum_i (x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T$$

Each point contributes to the class-specific deviation from its own mean.

✓ $\phi = m/n$; μ_k = class- k sample mean; Σ = pooled sample covariance.

Q14. Precision, Recall & F1 Score

Classifier A: precision=0.80, recall=0.60. Classifier B: precision=0.65, recall=0.85. (a) Compute F1. (b) Which is better for medical diagnosis?

■ Formula

$$F1 = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$$

■ (a) Classifier A

$$F1_A = 2 \cdot (0.80 \cdot 0.60) / (0.80 + 0.60) = 2 \cdot 0.48 / 1.40 \approx 0.686$$

■ (a) Classifier B

$$F1_B = 2 \cdot (0.65 \cdot 0.85) / (0.65 + 0.85) = 2 \cdot 0.5525 / 1.50 \approx 0.737$$

■ (b) Medical Diagnosis

In medical diagnosis, a false negative (missing a positive case, e.g., failing to detect a disease) is far more costly than a false positive.

Recall = TP/(TP+FN) measures the fraction of true positives caught. Classifier B has recall 0.85 vs A's 0.60 — it misses far fewer positive cases.

Classifier B is preferred despite lower precision (more false alarms are acceptable; missing disease is not).

✓ **$F1_A \approx 0.686$, $F1_B \approx 0.737$. Classifier B preferred — higher recall (0.85) minimises missed diagnoses.**

Q15. Logistic Regression — Gradient Descent Update Rule

State the gradient descent update rule for logistic regression (no regularisation). Define all symbols. Why prefer cross-entropy over squared error?

■ Update Rule

$$\theta_j \leftarrow \theta_j - \alpha \cdot (\partial J / \partial \theta_j)$$

Substituting the gradient of cross-entropy loss:

$$\theta_j \leftarrow \theta_j - (\alpha/n) \cdot \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$

■ Symbol Definitions

- θ_j : j-th parameter (weight)
- α : learning rate (step size)
- n : number of training examples
- $h_{\theta}(x) = \sigma(\theta^T x)$: sigmoid prediction, $\sigma(z) = 1/(1+e^{-z})$
- $y^{(i)} \in \{0,1\}$: true label
- $x_j^{(i)}$: j-th feature of i-th example

■ Why Cross-Entropy over Squared Error?

1. Non-convexity: squared error with sigmoid produces a non-convex loss with many local minima. Cross-entropy with sigmoid yields a convex loss with a unique global minimum.
2. Gradient vanishing: squared error gradient contains $\sigma'(z) = \sigma(z)(1-\sigma(z))$ which $\rightarrow 0$ for large $|z|$, slowing learning badly. Cross-entropy gradient simplifies to $(h-y)x$, with no vanishing sigmoid factor.
3. Probabilistic interpretation: cross-entropy is the negative log-likelihood under the Bernoulli model, making it the principled MLE objective.

✓ **Update rule above. Cross-entropy preferred for convexity, clean gradients, and MLE interpretation.**

Q16. PCA — Choosing k

Describe two criteria to choose k in PCA, and one practical reason to choose smaller k.

■ Criterion 1 — Explained Variance Threshold

Choose the smallest k such that:

$$\sum_{j=1}^k \lambda_j / \sum_{j=1}^d \lambda_j \geq \text{threshold (e.g. 95\%)}$$

This is the most common approach and directly answers 'how much information is retained?'

■ Criterion 2 — Scree Plot / Elbow

Plot eigenvalues λ_j in descending order. Choose k at the 'elbow' — the point after which each additional component explains much less variance. The elbow signals where noise components begin to dominate.

■ Practical Reason to Choose Smaller k

Computational / storage cost: retaining fewer components means smaller matrices and faster downstream models. In high-stakes deployability scenarios (mobile, embedded), a model that is 5% less accurate but 10× smaller may be strongly preferred.

Also: overfitting risk — retaining too many PC's (especially noisy ones) can degrade generalisation of the downstream classifier.

✓ **Criteria: (1) cumulative variance $\geq$ threshold; (2) scree plot elbow. Go smaller for compute/storage efficiency or to avoid noisy components.**

Q17. GDA vs Logistic Regression — Asymptotic Analysis

If GDA assumptions hold exactly, which achieves lower asymptotic error? What happens when they do not hold?

■ Key Relationship

When GDA's Gaussian class-conditional assumption holds, the posterior $P(y|x)$ derived from GDA is exactly logistic:

$$P(y=1|x) = \sigma(\theta^T x) \text{ for some } \theta$$

So GDA and logistic regression learn the same model family. But they use different information to fit it.

■ When Assumptions Hold Exactly

GDA uses the Gaussian structure as an additional constraint, effectively using more statistical information. This gives GDA lower variance and better performance with finite n . Asymptotically ($n \rightarrow \infty$), both converge to the Bayes-optimal boundary.

GDA is more data efficient — it needs fewer examples for equivalent performance.

■ When Assumptions Do Not Hold

GDA optimises the wrong objective (it fits a misspecified generative model). Its θ estimates are biased. Logistic regression makes no assumption about $P(x)$, so it still converges to the best linear decision boundary. In practice, LR is more robust to model misspecification.

✓ **If Gaussian: GDA wins with small n (more efficient). Asymptotically both equal. If non-Gaussian: LR wins (robust to misspecification).**

Q18. Bootstrap Sampling & Out-of-Bag Estimates

$n=6$ points, bootstrap sample of size 6. (a) $P(\text{specific point excluded})$? (b) Fraction excluded on average? What are these points called?

■ (a) $P(x^{(i)} \text{ not in bootstrap sample})$

Probability of not being selected in one draw = $(n-1)/n = 5/6$.

In $n=6$ independent draws with replacement:

$$P(x^{(i)} \text{ not selected}) = (1 - 1/n)^n = (5/6)^6 \approx 0.335$$

■ (b) Fraction Excluded on Average

As $n \rightarrow \infty$:

$$(1 - 1/n)^n \rightarrow e^{-1} \approx 0.368$$

So approximately 36.8% of the training set is excluded from each bootstrap sample on average. For $n=6$ specifically: $(5/6)^6 \approx 33.5\%$.

■ Out-of-Bag (OOB) Examples

Excluded points are called out-of-bag (OOB) examples. They can be used as a free validation set — the model trained on one bootstrap sample can be evaluated on its OOB examples without needing a separate held-out set. This is the OOB error estimate, commonly used in Random Forests.

✓ **(a) $(5/6)^6 \approx 0.335$. (b) $\approx 36.8\%$ excluded $\rightarrow$ called OOB examples $\rightarrow$ used as built-in validation for error estimation in Random Forests.**

Q19. Regression Tree — SSE Calculation

Points: (1,3),(2,5),(3,4),(8,9),(9,11). Split at $x=5$. Compute SSE before and after split.

■ SSE Before Split (all 5 points)

$$\bar{x} = (3+5+4+9+11)/5 = 32/5 = 6.4$$

$$\text{SSE} = (3-6.4)^2 + (5-6.4)^2 + (4-6.4)^2 + (9-6.4)^2 + (11-6.4)^2$$

$$= 11.56 + 1.96 + 5.76 + 6.76 + 21.16 = 47.2$$

■ SSE After Split at $x=5$

Left ($x \leq 5$): {(1,3),(2,5),(3,4)} $\rightarrow \bar{x}_L = 12/3 = 4$

$$\text{SSE}_L = (3-4)^2 + (5-4)^2 + (4-4)^2 = 1 + 1 + 0 = 2$$

Right ($x > 5$): {(8,9),(9,11)} $\rightarrow \bar{x}_R = 20/2 = 10$

$$\text{SSE}_R = (9-10)^2 + (11-10)^2 = 1 + 1 = 2$$

$$\text{SSE}_{\text{split}} = \text{SSE}_L + \text{SSE}_R = 2 + 2 = 4$$

■ Comparison

$$\text{SSE reduction} = 47.2 - 4 = 43.2 \text{ (91.5\% reduction!)}$$

✓ SSE before = 47.2, SSE after split = 4. Excellent split — 91.5% SSE reduction by separating the two natural groups.

Q20. Naïve Bayes — Conditional Independence Assumption

Give an example where the Naïve Bayes independence assumption is violated. Why does NB still often perform well?

■ Violated Assumption

Naïve Bayes assumes: $P(x_1, x_2, \dots, x_d | y) = \prod_j P(x_j | y)$, i.e. features are conditionally independent given the class.

■ Realistic Violation Example

Spam email classification: the words 'free' and 'money' are positively correlated — if 'free' appears, 'money' is more likely to appear too. Emails advertising free money contain both together far more often than independence would predict. The model double-counts correlated evidence.

Another example: in document classification, 'New' and 'York' are highly dependent — knowing 'New' dramatically increases $P(\text{'York'})$.

■ Why NB Works Anyway

For classification we only need the argmax of $P(y|x)$, not a calibrated probability. Even if the absolute probabilities are wrong, the class with the highest (miscalibrated) posterior is often still the correct class. The decision boundary can still be good despite the bad probability estimates.

Formally: NB classification error can be small even when the probability estimates are biased, as long as the rankings are preserved.

✓ Example: 'free' and 'money' in spam are correlated. NB still works because classification only needs correct argmax, not calibrated probabilities.

■ What to Revise Before the Exam

● Logistic Regression

- Sigmoid function and its derivative $\sigma'(z) = \sigma(z)(1-\sigma(z))$
- Cross-entropy loss and its gradient — know the derivation
- Regularisation (L1/L2): how it modifies the update rule and prevents overfitting
- Multi-class extension: Softmax regression
- Feature scaling importance for gradient descent convergence

● Naïve Bayes

- Bernoulli vs Multinomial NB — which to use for which data?
- Laplace smoothing formula and effect on probability estimates
- Log-space computation to avoid numerical underflow
- Text classification pipeline: tokenisation, vocabulary, feature encoding

● Gaussian Discriminant Analysis

- Full derivation of MLE for ϕ , μ_0 , μ_1 , Σ
- How shared vs per-class Σ changes the decision boundary (linear vs quadratic)
- Relationship to LR: when GDA collapses to logistic form
- Asymptotic comparison and bias-variance trade-off vs LR

● PCA

- Covariance matrix: how to compute and interpret
- Eigenvector decomposition: know what eigenvectors and eigenvalues represent
- Explained variance ratio and cumulative explained variance
- Whitening / data pre-processing before PCA
- PCA limitations: linear only, sensitive to outliers, assumes Gaussian

● K-Means Clustering

- Full derivation of why WCSS is non-increasing in K
- K-Means++ initialisation algorithm (know the sampling distribution)
- Time complexity: $O(nKdt)$ — n points, K clusters, d dimensions, t iterations
- How to evaluate clustering quality: silhouette score, Davies-Bouldin index

● Decision Trees

- Entropy (information gain) vs Gini impurity — pros/cons

- CART algorithm for regression trees (SSE) vs classification trees
- All pruning methods: pre-pruning parameters, reduced-error, cost-complexity
- Random Forests: bagging + feature subsampling, out-of-bag error

● Metrics & Model Evaluation

- Confusion matrix: TP, TN, FP, FN — derive all metrics from it
- ROC curve construction (vary threshold, plot TPR vs FPR)
- Precision-Recall curve — better for imbalanced data than ROC
- Bias-variance decomposition and its relationship to under/overfitting
- Cross-validation: k-fold, stratified k-fold, leave-one-out

● Bootstrap & Ensemble Methods

- Bootstrap sampling maths: $(1 - 1/n)^n \rightarrow e^{-1} \approx 0.368$
- Bagging (Bootstrap Aggregating): reduces variance
- Boosting (AdaBoost, Gradient Boosting): reduces bias, sequential learners
- Stacking: meta-learner combines base model predictions

Study Tip: For each topic above, practise deriving the formulas from scratch, not just recognising them. Exam questions often ask you to 'derive' or 'show that'. Also practise explaining the intuition in plain English — many marks come from conceptual understanding, not just computation.